

$$\begin{aligned}
& \wedge tx_commit_time[t1] \leq tx_snapshot[t2] \\
& \vee \wedge (t2 = InitTx \vee Workload[t2].level = \text{"SER"}) \\
& \wedge \langle t1, t2 \rangle \in Derived_AR \\
& \} \\
OpsOf(t) & \triangleq \text{DOMAIN } op_seq[t] \\
WriteSet(t) & \triangleq \\
& \text{IF } t = InitTx \\
& \text{THEN } Keys \\
& \text{ELSE } \{op_seq[t][i].key : i \in \{j \in OpsOf(t) : op_seq[t][j].type = \text{"W"}\}\} \\
WriteConflict(t1, t2) & \triangleq WriteSet(t1) \cap WriteSet(t2) \neq \{\} \\
LastWriteIndex(t, x) & \triangleq \\
& \text{IF } t = InitTx \\
& \text{THEN } 1 \\
& \text{ELSE LET } xw \triangleq \\
& \quad \{i \in OpsOf(t) : op_seq[t][i].key = x \wedge op_seq[t][i].type = \text{"W"}\} \\
& \quad \text{IN IF } xw = \{\} \text{ THEN } 0 \text{ ELSE CHOOSE } max_i \in xw : \forall i \in xw : max_i \geq i \\
FirstOpIndex(t, x) & \triangleq \\
& \text{IF } t = InitTx \\
& \text{THEN } 1 \\
& \text{ELSE LET } xo \triangleq \{i \in OpsOf(t) : op_seq[t][i].key = x\} \\
& \quad \text{IN IF } xo = \{\} \text{ THEN } 0 \text{ ELSE CHOOSE } min_i \in xo : \forall i \in xo : min_i \leq i \\
T_writes(t, x, v) & \triangleq \\
& \text{IF } t = InitTx \\
& \text{THEN } v = InitVal \\
& \text{ELSE } LastWriteIndex(t, x) > 0 \wedge op_seq[t][LastWriteIndex(t, x)].val = v \\
T_reads_ext(t, x, v) & \triangleq \\
& t \neq InitTx \wedge FirstOpIndex(t, x) > 0 \wedge \\
& \quad op_seq[t][FirstOpIndex(t, x)].type = \text{"R"} \wedge \\
& \quad op_seq[t][FirstOpIndex(t, x)].val = v \\
MaxAR(S, AR_rel) & \triangleq \\
& \text{CHOOSE } max_t \in S : \forall t \in S \setminus \{max_t\} : \langle t, max_t \rangle \in AR_rel \\
IntAxiom(t) & \triangleq \\
& \forall i \in OpsOf(t) : \\
& \quad op_seq[t][i].type = \text{"R"} \Rightarrow \\
& \quad \text{LET } prev_ops \triangleq \\
& \quad \quad \{j \in 1 \dots (i-1) : op_seq[t][j].key = op_seq[t][i].key\} \\
& \quad \text{IN } prev_ops \neq \{\} \Rightarrow \\
& \quad \quad \text{LET } max_j \triangleq \text{CHOOSE } j \in prev_ops : \forall k \in prev_ops : j \geq k \\
& \quad \quad \text{IN } op_seq[t][i].val = op_seq[t][max_j].val
\end{aligned}$$